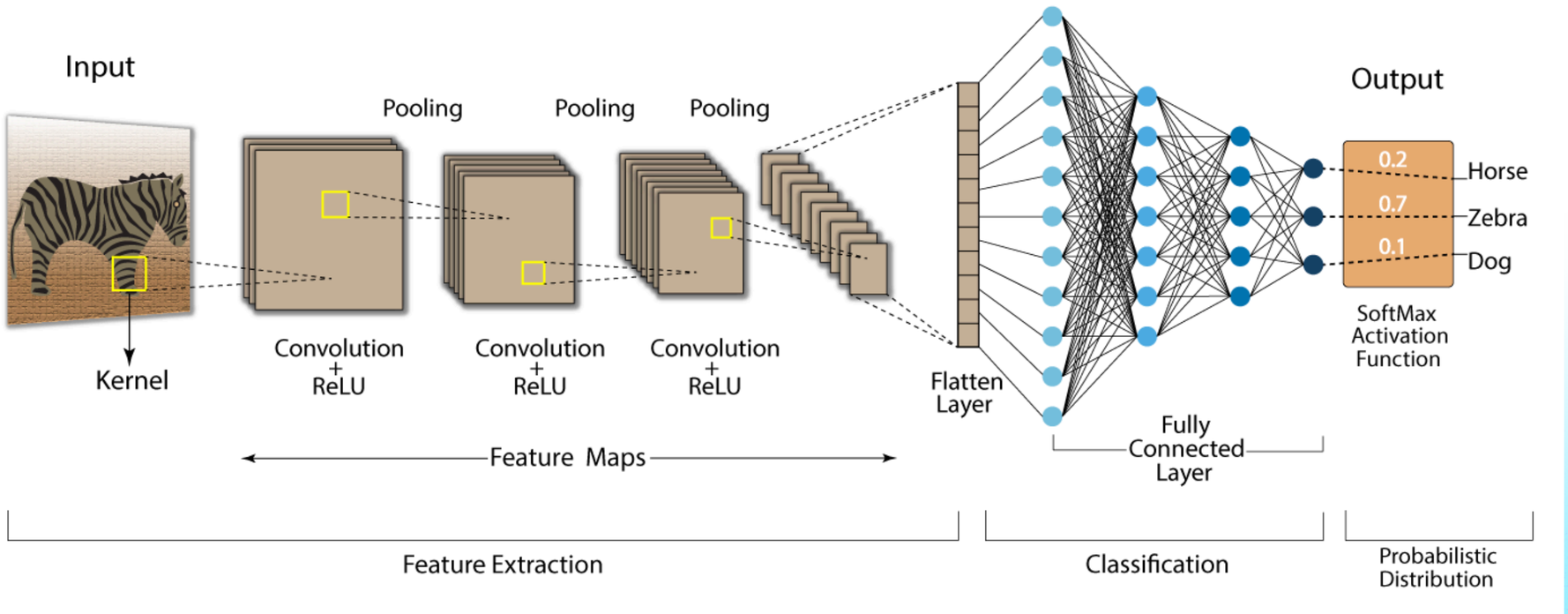
What does a CNN really see?

EdinburgAI workshop 4, sem1

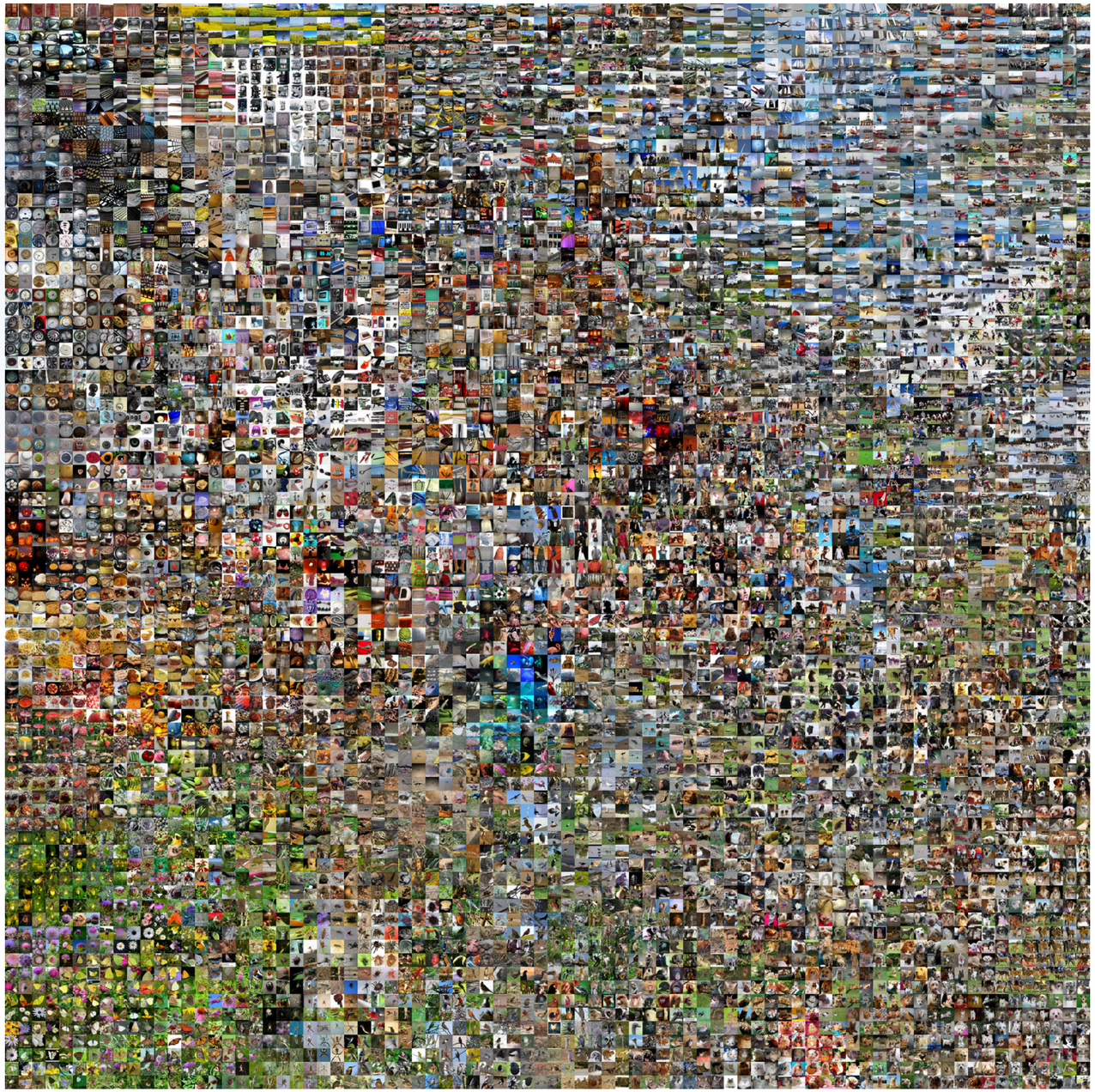
Convolutional Neural Networks



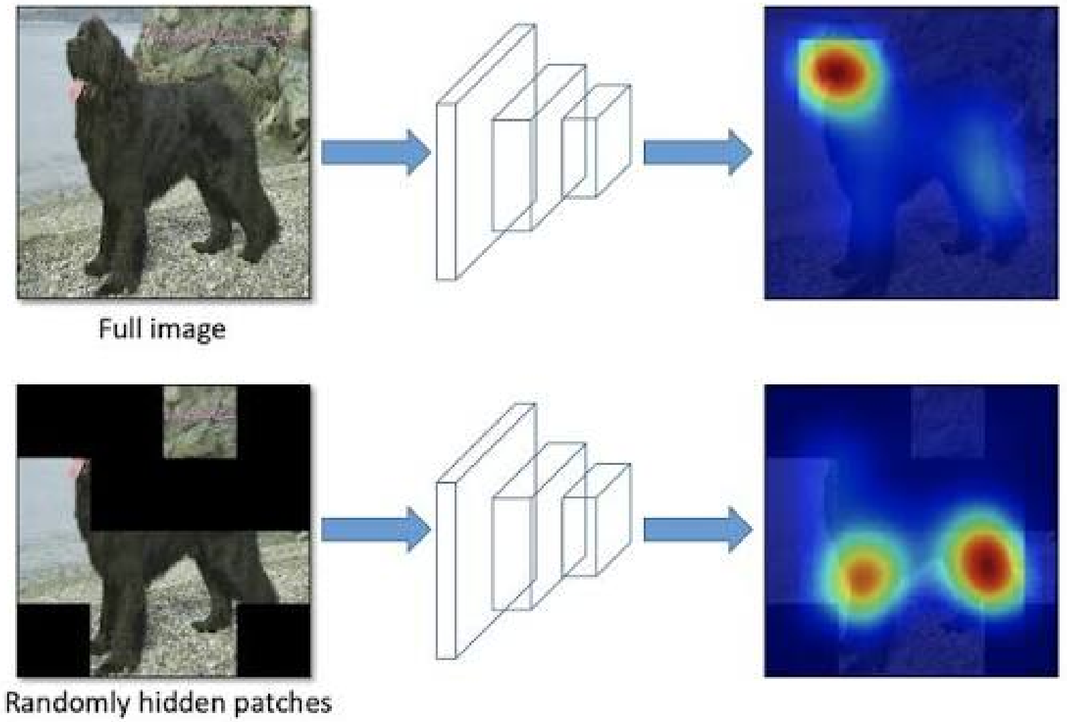
Let's see some examples!

Problems with images we use to train CNNs

How do we know what the model looks for in an image to classify it?



t-SNE dimensionality reduction



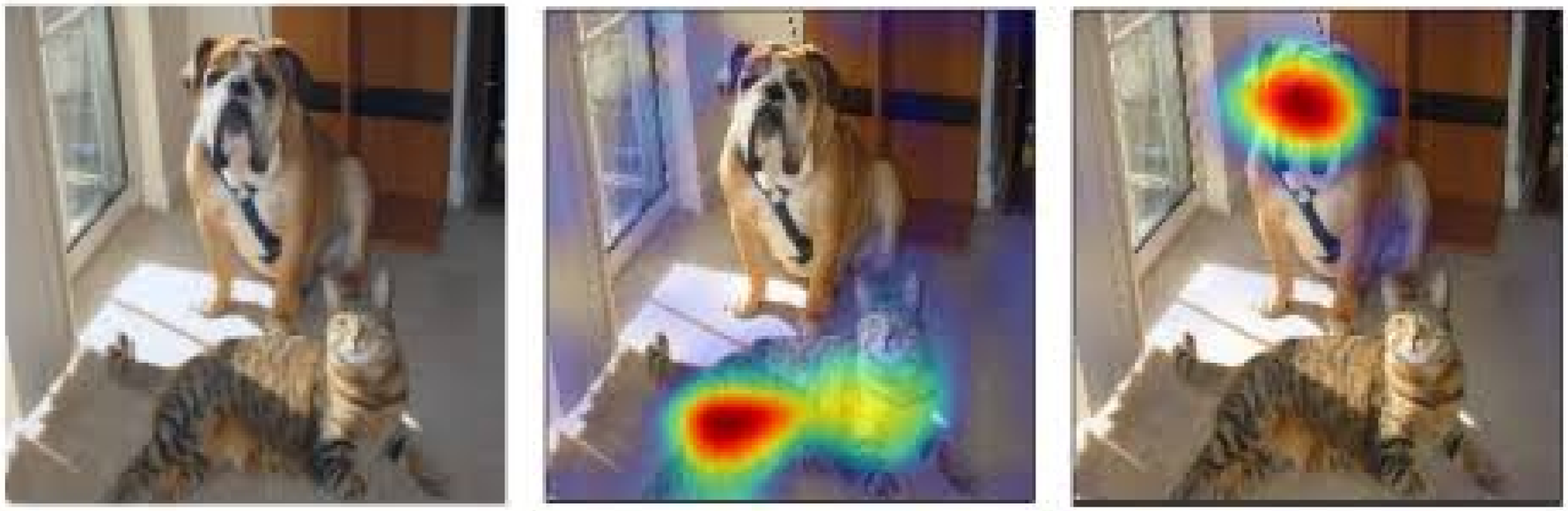
occlusion

Let's inspect the neural network itself!

Today's focus: backpropagation

GradCAM

For those who want to look a little further...



Original Image | Grad-CAM ‘Cat’ | Grad-CAM ‘Dog’

Useful Resources

https://cs.stanford.edu/people/karpathy/convnetjs/demo/cifar10.html

https://poloclub.github.io/cnn-explainer/

https://medium.com/analytics-vidhya/deep-dream-visualizing-the-features-learnt-by-convolutional-networks-in-pytorch-b7296ae3b7f